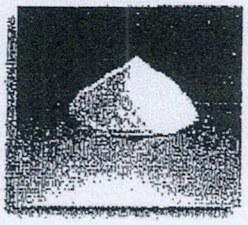
16.

TAIYO KOKO

SODIUM MOLYBDATE

(1) CHEMICAL SPECIFICATION



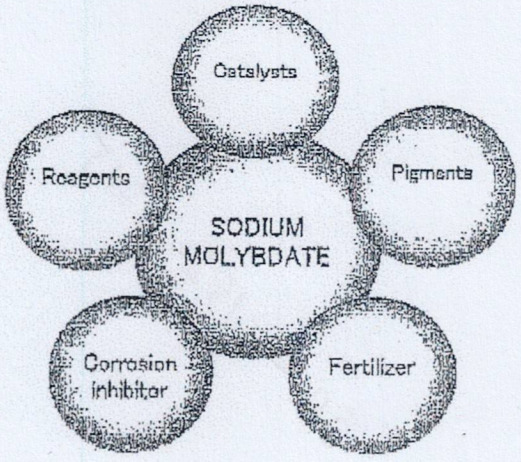
ELEMENT etc.

Solubility in water	To pass test
pH (5w/v% ($Na_2MoO_4 \cdot 2H_2O$), 25°C)	10.0 max.
SO_4	0.01 %max.
PO_4	0.001 %max.
SiO_2	0.005 %max.
Fe+Pb	0.003 %max.
$Na_2MoO_4 \cdot 2H_2O$	99.0 %min.

(2) PRODUCT DESCRIPTION

CHEMICAL FORMULA	$Na_2MoO_4 \cdot 2H_2O$
MOLECULAR WEIGHT	241.95
SPECIFIC GRAVITY	3.28
MELTING POINT	687°C

- A white crystalline powder. Soluble in 1.7 parts of cold water, about 0.9 part boiling water. It decomposes when its aqueous solution is acidified and heated. Soidum molybdate dihydrate forms an anhydrous salt, a colorless crystal, when heated to 100°C.



(3) APPLICATION

- Catalysts. Reagent for alkaloids. Fertilizer. Micronutrient for plants and animals. Corrosion inhibitor. Manufacture of inorganic and organic pigments.

(4) PACKAGING

- 25kgs. in a polyethylene bag and then packed in a triple layer paper bag.

Substances/Preparations which,in contact with water,emit flammable gases :

Not classified

Oxidizing Solids : Not classified

Corrosive to metals : Not classified

Health Hazards

Acute toxicity(oral,skin) : Category 5

- (inhalation/gas) : Not classified
- (inhalation/vapour) : Not available
- (inhalation/dust) : Category 4
- (inhalation/mist) : Not classified

Serious eye damage/Eye irritation : Category 2A

Germ cell mutagenicity(Heritable mutations in germ cell) : Category 2

Carcinogenicity : Category 2

Specific target organ systemic toxicity-single exposure (respiractory tract irritation) : Category 3

Specific target organ systemic toxicity-repeated exposure : Category 2

Hazadous to the aquatic environment(acute toxicity) : Not available

Hazadous to the aquatic environment(chronic toxicity) : Not available

GHS Labelling

Health Hazards & Environmental Effects Information

Harmful if swallowed

Eye irritation

Heritable mutagenicity

Carcinogenicity

Respiractory sensitization

Danger of serious damage to organs by repeated ,prolonged exposure

Harmful to aquatic organisms

Matters that require attention

Use personal protection equipment and exhaust ventilation and avoid exposure

First issue : May 24,2000

Revised : August 1,2007

MSDS No.M-002

MATERIAL SAFETY DATA SHEET

1. CHEMICAL PRODUCT AND COMPANY IDENTIFICATION

PRODUCT NAME : Sodium Molybdate

MSDS NUMBER : M-002

MANUFACTURER :

COMPANY NAME : Taiyo Koko Co.,Ltd.

ADDRESS : Head Office

1-1-39 Isobe-dori,Chuo-ku,Kobe,Hyogo 651-0084 Japan

Telephone No.： 81-78-231-3700 Fax No.： 81-78-252-3100

Ako Plant

1603-1 Aza-Higashioki,Nakahiro,Ako,Hyogo 678-0232 Japan

EMERGENCY TELEPHONE NUMBER

Taiyo Koko Co.,LTD. Ako Plant Quality Control Sect.

Telephone No.： 81-791-43-2250 Fax No.： 81-791-43-6949

2. COMPOSITION/INFORMATION ON INGREDIENTS

SUBSTANCE/PREPARATION : Substance

CEMICAL NAME : Sodium Molybdate,Dihydrate

INGREDIENTS AND COMPOSITION : Sodium Molybdate,Dihydrate 98.5% min.

CHEMICAL FORMULA : $Na_2MoO_4.2H_2O$

CAS REGISTER NUMBER : 10102-40-6

UN CLASS/UN No. : Not classified

3. HAZARDS IDENTIFICATION

GHS CLASSIFICATION :

Physical Hazards

Flammable Solids : Not classified

Self-reactive Substances/Preparations : Not classified

Pyrophoric Solids : Not classified

Self-heating Substances/Preparations : Not classified

- Use only in well-ventilated areas
- Do not breathe dust or fume
- Wash hands thoroughly after handling.
- If inhaled,immediately remove to fresh air.If not breathing,give artificial respiration. If breathing is difficult,give oxgen. Call a physician.
- In case of contact with eyes,rinse immediately with plenty of water and seek medical advice.
- If swallowed,rinse mouth immediately and call a physician.
- If feel unwell,seek medical advice immediately.
- Avoid release to the environment.
- Keep container tightly closed and store in moderately cool and dark place with locked.
- Disposal considerations : Comply with all fedral,state and local regulations.

4. FIRST-AID MEASURES

INHALATION : Removal of exposed individual from area to fresh air. If not breathing ,give artificial respiration.If breathing is difficult,give oxgen. Call a physician.

EYE CONTACT : Immediately flush eyes with plenty of water. Call a physician.

SKIN CONTACT : Immediately flush skin with plenty of water using a soap while removing contaminated clothing and shoes. Call a physician.

INGESTION : Rinse mouth with water. Call a physician.

5. FIRE FIGHTING MEASURES

EXTINGUISHING MEDIA : Any kind of extinguishing media available like foam,powder,carbon dioxide ,water and dry sand.

FIREFIGHTINGINSTRUCTIONS : Harmful fume(sodium oxide etc) will form with heating. Reaction with halogen causes explosion

Keep personnel removed from and upwind of fire.

Wear full fire-fighting turn-out gear and respiratory protection.

Cool container with water spray.

6. ACCIDENTAL RELEASE MEASURES :

- Isolate the hazard area and deny entry to unnecessary and unprotected personnel

▯ Vacuum or sweep material and place in a disposal container.

▯ This material is a water pollutant and should be prevented from contaminating soil or from entering sewage and drainage systems and bodies of water.

▯ For personal protective equipment,refer to 8.EXPORE CONTROLS,PERSONAL PROTECTION.

7. HANDLING AND STORAGE :

HANDLING : Do not breathe dust. Avoid contact with skin,eyes or clothing.

Minimize dust generation and accumulation.

Refer ENGINEERING CONTROLS and PERSONAL PROTECTIVE EQIUPMENT in

"8.EXPORE CONTROL,PERSONAL PROTECTION"

8. EXPOSURE CONTROL,PERSONAL PROTECTION

EXPOSURE GUIDELINES : ACGIH(2005)

TLV-TWA 0.5mg/m3(water soluble compound: as Mo) : A3

ENGINEERING CONTROLS : Handle in fully enclosed systems and equipments or ensure local exhaust ventilation.

PERSONAL PROTECTIVE EQUIPMENT :

RESPIRATORY PROTECTION : Recommend to use filter respirator

EYE PROTECTION : Recommend to wear safety glasses

HAND,SKIN AND BODY PROTECTION : To avoid any contact,recommend to wear clothing such as rubber gloves,body-cover suits

9. PHYSICAL AND CHEMICAL PROPERTIES

APPEARANCE : White crystallized powder

ODOR : Odorless

pH : Not available

FREEZING/MELTING POINT : 687°C(mp)

BOILING POINT : Not available

FLASH POINT : Not available

FLAMMABILITY(SOLID,GAS) : Not available

FLAMMABILITY or EXPLOSIVE LIMIT : Not available

VAPOR PRESSURE : Not available

VAPOR DENSITY : Not available

DENSITY(SPECIFIC GRAVITY) : 3.28

SOLUBILITY IN WATER : Soluble 45.6%(100 °C) 39.3%(15.5 °C)

SOLUBILITY IN ALCOHOL : Not available

OCTANOL/WATER PARTITION COEFFICIENT : Not available

DECOMPOSITION POINT : 100 °C

10. STABILITY AND REACTIVITY

CHEMICAL STABILITY : Stable under normal temperature & pressure conditions

CONDITION TO AVOID : Heat and sunshine

INCOMPATIBILITY WITH OTHER MATERIALS : melt magnesium, halogen

HADARDOUS DECOMPOSITION PRODUCTS : Harmful fume(sodium oxide etc) will form when heated

11. TOXICOLOGICAL INFORMATION

ACUTE TOXICITY : Oral(rat) LD50 4000mg/kg

: Skin(rat) LD50 >2000mg/kg

: inhalation(rat) LD50 2.08mg/L

CORROSIVE AND IRRITANT PROPERTIES : Reportedly recognition of irritation when exposed

SERIOUS EYE DAMAGE/EYE IRRITATION : Reportedly recognition of irritation when exposed

RESPIRATORY SESITIZATION/SKIN SENSITIZATION : Not available

GERM CELL MUTAGENICITY(HERITABLE MUTATIONS IN GERM CELL) : In vitro mouse test:positive

CARCINOGENICITY : ACGIH(2005),water soluble Mo-compound A3

REPRODUCTIVE AND DEVELOPMENT TOXICITY : in vivo rat test : positive

SPECIFIC TARGET ORGAN SYSTEMIC TOXICITY-SINGLE EXPOSURE :

Recognition of irritation on respiratory organs & conjunctiva when exposured.

Animal testing indicates recognition of irritation on respiratory organs & conjunctiva when exposed by inhalation.

SPECIFIC TARGET ORGAN SYSTEMIC TOXICITY-REPEATED EXPOSURE :

Animal testing of repeated exposure-inhalation shows recognition of restraint on gaining weight & hematoma of lungs.

Repeated oral injestion indicates deformed bones & joints and fur fading.

RESPIRATORY INHALATION EFFECTS : Not available

12. ECOLOGICAL INFORMATION

ECOTOXICITY :

Fish : rainbow trout LC50(96hr)=800mg/L

: silver salmon LC50(96hr)=>1000mg/L

Others(Crustacea,Algae) : Not available

BIODEGRADABILITY : Not available

BIOACCUMULATION : Not available

OTHER HAZADOUS EFFECTS : Not available

13. DISPOSAL CONSIDERATIONS

Comply with all applicable fedral,state and local regulations

14. TRANSPORTATION INFORMATION

UN CLASS/UN NO. : Not classified

Follow all regulations in user's country and/or region

15. REGULATORY INFORMATION

Ensure this material in compliance with federal requirements and ensure conformity to local laws and regulations.(Regulatory information withregard to this substance in user's country or region should be examined by user's own responsibility).

16. OTHER INFORMATION

This information is furnished without warranty,express or implied,except that it is accurate to the best knowledge of Taiyo Koko Co.,Ltd. IT relates only to the specific material designated herein,and does not relate to use in combination with any other material or in any process.Taiyo Koko Co.,Ltd. assumes no legal responsibility for use of or reliance upon this information.

Processing Flow Sheet

Manufacturer	Taiyo Koko Co.,Ltd.
Commodity	Sodium Molybdate

